



Md Raihan Patwary

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Address: Railway Housing Society, Akbarshah, Pahartali, 83, Road-6, 4202, Chattogram, Bangladesh (Home)

Address: Shaheed Shamsul Haque Hall, Bangladesh Agricultural University, Room No: 210/A, 2202, Mymensingh, Bangladesh (Work)

ABOUT ME

MS researcher in Animal Nutrition at Bangladesh Agricultural University, with hands-on experience in animal nutrition, insect-based alternative proteins, and livestock research. My work explores sustainable feed systems and the bioconversion potential of insects such as black soldier fly, mealworm, and superworm, including their waste-degradation capacity and how to optimize their production, using machine learning and chemometrics to model and predict nutritional outcomes

EDUCATION AND TRAINING

01/12/2024 – CURRENT Mymensingh, Bangladesh

MASTER OF SCIENCE Bangladesh Agricultural University, Department of Animal Nutrition

Website <https://an.bau.edu.bd/> | **Field of study** Animal Nutrition

06/01/2019 – 07/11/2024 Mymensingh, Bangladesh

BACHELOR OF SCIENCE Bangladesh Agricultural University

Ruminant and Non-ruminant Nutrition, Biochemistry, Physiology, Molecular Genetics, Environmental Science, Dairy and Beef Cattle Production, Reproduction of Farm Animals, Reproductive Biotechnology, Dairy Chemistry, Dairy Microbiology, Statistics.

Website <https://ah.bau.edu.bd/> | **Field of study** Animal Husbandry | **Final grade** 3.426 out of 4.00 |

Type of credits Uniform Recommended Grading System | **Number of credits** 192

Chittagong, Bangladesh

HIGHER SECONDARY CERTIFICATE (HSC) Chittagong College

Website <https://ctgcollege.gov.bd/> | **Field of study** Intermediate of Science | **Final grade** 5.00 out of 5.00

WORK EXPERIENCE

RESEARCH ASSISTANT – DEPARTMENT OF ANIMAL NUTRITION, BAU – 01/12/2024 – Current – MYMENSINGH, BANGLADESH

Monitor larval development and feeding patterns of mealworms | Conduct wet lab analyses | Support team projects | Apply Python and R for predictive modeling and statistical analysis | Review literature and assist in manuscript preparation

INTERN QUALITY CONTROL OFFICER – FEED MILL – NAHAR AGRO – 12/06/2024 – 24/07/2024

Active participation in receiving of raw materials | Ration formulation | NIR analysis of raw and final products

INTERN – NAHAR AGRO – 25/07/2024 – 30/09/2024 – CHITTAGONG, BANGLADESH

Website: <https://naharagro.com/>

Maintenance of DOC(Day old chicks) | Management of semi automated poultry shed | Feed formulation

INTERN – DEPARTMENT OF LIVESTOCK SERVICES (DLS) – 05/2024 – 06/2024 – CHATTOGRAM, BANGLADESH

Website: <https://dls.gov.bd/>

PROJECTS

23/04/2026 – CURRENT
Study of black soldier fly, superworm, and yellow mealworm as alternative-protein candidates, improving nutrient content while quantifying waste degradation efficiency, with ML-augmented LCA.

Starting my journey with black soldier flies, applying flaxseed oil to increase their overall omega-3 fatty acid content, my team now aims to use rice straw, an agricultural by-product available in large amounts, to find an economical and nutritionally effective approach.

01/11/2024 – 13/02/2026
Life Cycle Assessment of Yellow Mealworm (*Tenebrio molitor*) Larvae in Bangladesh: Prospects for Commercialization as a Sustainable Poultry Protein Source

I am working on this project as the project leader. I reduced the production cost by 27% and used machine learning models to predict the LCA. I was awarded the National Science and Technology (NST) Fellowship.

01/04/2025 – 13/10/2025
Omega-3 Enriched Broiler Production

This project is funded my The World bank. Helping the peers and conducting feeding trials. Active participation in data collection, lab analysis and statistical analysis.

LANGUAGE SKILLS

Mother tongue(s): **BENGALI**
Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C2	C2	C2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

VOLUNTEERING

Mymensingh
Examiner In Bangladesh Biology Olympiad

02/12/2021 – 12/2022 Bangladesh Agricultural University Career Club
Research and Development Manager

PUBLICATIONS

2025
[Roughage and concentrate based feeding and their impact on beef quality](#)

Patwary, M., Rahman, M., Yousuf, S., Mazumder, A., Sojib, M., & Murshed, H. (2025). Roughage and concentrate based feeding and their impact on beef quality – A review. Meat Research, 5(5). <https://doi.org/10.55002/mr.5.5.125>

Journal Name: Meat Research Journal | Volume, Issue and Pages: Vol. 5 No. 5 (2025): October

2026
Comparing chemometric and machine learning approaches for predicting yellow mealworm larval weight from proximate composition

In final preparation.

HONOURS AND AWARDS

13/01/2026
National Science and Technology (NST) Fellowship – Ministry of Science and Technology

The NST Fellowship is awarded to young scientists to encourage and support their research journey. I was awarded this fellowship for my project on mealworms.

2023
Second Runners Up – HULT Prize at BAU

I was the team leader.

2022
Second Runners Up – HULT Prize at BAU

Participation in the regional round in Mumbai (online).

2016
SECONDARY SCHOOL CERTIFICATE (SSC), Education Board Scholarship – Chittagong Board

2013
Junior School Certificate Exam, Education Board Scholarship – Chittagong Board

● **SKILLS**

Microsoft Office

Programing

R | Python

Data Analysis and Visualization

NIR | Minitab | Microsoft Excel | SPSS

Lab Skills

PCR | In-Vitro Analysis | Antioxidant Profiling | ELISA | Proximate Analysis

Others

LinkSquare | Zotero

● **CERTIFICATIONS**

Coursera
Introduction to Genomic Technologies

Mode of learning: Online

Link https://drive.google.com/file/d/17SyyREXMXcbU0ay9Tf2Pbr6lZSk8WFOe/view?usp=drive_link

Kaggle
Intermediate Machine Learning

Mode of learning: Online

Link https://drive.google.com/file/d/14lvxFkXvFi9qDPl1nn8vp_UQcmdZBXlr/view

Cognitive Class
R for Data Science

Mode of learning: Project based

Link <https://drive.google.com/file/d/1jSHyOhTJK-ktCnLvX8avs4eHlReGa0Ai/view>

Angkuron
Data Analysis and Visualization using R

Mode of learning: Online

Link https://drive.google.com/file/d/10bNtV5_mnliHPNsWvZMLP4TrRj46tna/view

● **CONFERENCES AND SEMINARS**

02/2025 Dhaka, Bangladesh
13th International Poultry Conference

Plagiarism Check and Manuscript Submission Process, Bangladesh Agricultural university Research Club.

● NETWORKS AND MEMBERSHIPS

02/02/2026 – CURRENT Bangladesh
World's Poultry Science Association, Bangladesh (WPSA-BB)

● RECOMMENDATIONS

Dr. Md Safiqur Rahaman Shishir Associate Professor

Current department head of Department of Animal Nutrition. MS supervisor and project conductor.

Email shishir.an@bau.edu.bd | Phone (+880) 1717817715

Dr. Tarikul Islam Associate Professor

Faculty Member, Department of Entomology. MS co-supervisor and principal investigator.

Email tarikul.entom@bau.edu.bd | Phone (+880) 1721204164

● SPECIALIZED TRAINING AND WORKSHOP

27/05/2026
Online workshop on "Love cage handling of Black Soldier Flies"

Organized by the Department of Animal Nutrition, BAU. I was the **lead instructor** and showcased to a small group of farmers how they can cost-efficiently design a love cage for collecting the eggs of BSFs.

03/07/2025 – 06/07/2025
Hands on Training on " Advanced Next Generation Sequencing Data Analysis (NGS) "from Dawn of Bioinformatics Limited and Department of Microbiology & Hygiene, BAU

02/11/2024 – 12/11/2024
Training on " Data Analysis: R & SPSS " from Graduate Teaching Institute,BAU
